BeeQ: Quantum Fidelity Kernels and the Limits of Transfer in Acute Honey-Bee Toxicity Prediction

Anonymous authors

***Abstract*—BeeQ compares quantum and classical molecular kernels and examines toxic-compound recovery beyond Development. Six model families use 893 curated molecules, ten fixed descriptors (X10), and nested molecular-cluster-disjoint validation with five outer and four inner folds. Random Forest has the highest mean Development AUROC (0.748 ± 0.057). Product and IQP-ZZ show small, inconsistent differences from RBF and highly aligned Gram matrices (CKA 0.982 and 0.985); Product's effective rank closely matches an RBF control at its small-angle bandwidth. Transfer is limited: original historical-holdout AUROCs span 0.462–0.706, while sensitivity is 0.082–0.265 at Development-selected thresholds. Retrospective balancing of the originally unweighted MLP yields Development AUROC 0.688 with fixed hyperparameters and 0.679 with repeated selection, versus 0.677 originally. Holdout toxic recovery rises from 4/49 to 11/49 and 13/49, but false positives rise from 4 to 16 and 22. Both balanced variants recover one previously missed toxic CR8 compound and add a false positive; CR8 contains only two toxic compounds. The historical holdout was previously inspected, and the additional evaluations are retrospective. Neither a robust quantum predictive advantage nor reliable toxic recovery is established for the tested X10 protocol. Richer representations and kernel adaptation remain untested.**

***Index Terms*—quantum kernels, honey-bee toxicity, molecular validation, class imbalance, applicability domain, generalization**

I. Introduction

Quantum feature maps encode classical inputs as quantum states, whose overlaps define kernels for a classical learning algorithm [1], [2]. A change of feature map can therefore change the similarity geometry used by a support vector classifier (SVC), without replacing its optimization procedure. Whether that change improves prediction depends on the representation, data distribution, and classical comparator. Geometric-difference theory links kernel geometry to possible differences in learning performance [3]; fidelity kernels can also exhibit exponential concentration under specific conditions [4]. These observations motivate controlled comparisons, rather than an assumption that quantum encoding itself improves molecular prediction [5].

Acute toxicity to *Apis mellifera* provides an environmentally relevant setting, given the role of pollinators in crop production [6]. Earlier work used graph-kernel SVMs for pesticide toxicity [7]. ApisTox subsequently integrated ECOTOX, PPDB, and BPDB into a curated benchmark [8]; a recent study evaluated fingerprints, graph kernels, graph neural networks (GNNs), and pretrained representations [9]. Chemical similarity across training and evaluation can inflate performance, making molecular separation central to QSAR validation [10], [11], [12].

BeeQ addresses a narrower question than a search for the best toxicity predictor: on a fixed, ten-descriptor representation, do Product and IQP-ZZ fidelity kernels yield a persistent predictive change or a distinct similarity geometry relative to RBF? X10 preserves a common input budget across classical models and ten-qubit maps; it is not an empirically optimized representation. We distinguish a predictive comparison using independently selected bandwidths from a post hoc RBF control at Product's analytically implied bandwidth. We also examine a second, practically important question: how well are toxic compounds recovered beyond Development? The previously inspected historical holdout and the eight-compound Costa Rican challenge panel (CR8) have different provenance and different inferential limits. An additional MLP balancing analysis addresses a training asymmetry in the original comparison without replacing its experimental history.

II. Methods

A. Dataset, Endpoint, and Fixed Representation

BeeQ893 derives from the 1,035 ApisTox compounds [8]. The recorded curation and structural-standardization stage, including exclusion of metal-containing structures and treatment of salts and charge states with RDKit, retained 893 organic molecules [13], [14], [15]. The final versioned tables contain 622 class-0 and 271 class-1 compounds. We retain the ApisTox binary endpoint: class 1 when the selected acute $\mathrm{LD}_{50} \leq 11\,\mu$g/bee, and class 0 otherwise [8], [16]. This threshold defines the endpoint and was not fitted to model performance. Labels summarize the most severe available compound-level acute toxicity across the source exposure routes; heterogeneous assays and routes are not made homogeneous by curation. The task is acute-hazard classification, rather than continuous LD_{50} regression or prediction of field risk.

Every model uses the same ordered X10 vector: MolLogP, MolWt, TPSA_SP, NumHDonors, NumRotatableBonds, NumAromaticRings, nHalogen, n_OP, LiPHEX_prediction, and sasa002_frac_polar_hetero_only. MolLogP is the Wildman–Crippen octanol/water estimate [17]; LiPHEX supplies a frozen estimate of hexadecane/water partitioning [18]. TPSA includes sulfur and phosphorus, nHalogen counts F/Cl/Br/I, and n_OP uses the frozen organophosphorus SMARTS pattern specified in Supplementary Table D1. The final descriptor is a supplied fraction of accessible surface associated with polar heteroatoms. The ten dimensions support one input per qubit, with chemically informed descriptors, but were not selected by a predictive feature-ranking procedure. All descriptor values, molecules, labels, and partitions remain fixed in the present

revision. Supplementary Table D1 provides the complete definitions and distinguishes reproducibility from frozen tables from regeneration of the two supplied in-house descriptors.

B. Molecular Groups and Nested Validation

Morgan fingerprints (radius 2, 1,024 bits, chirality enabled) define Tanimoto similarities [19], [13]. Butina clustering uses distance $d = 1 - T$ and cutoff 0.60 [20], corresponding to $T \geq 0.40$ for a molecule's assignment relative to a cluster center. This does not impose a universal upper bound on similarities between different clusters. The frozen corpus has 462 clusters, including 334 singletons and a maximum cluster size of 40. Development contains 712 molecules (490 non-toxic, 222 toxic; 374 clusters); the historical holdout contains 181 (132 non-toxic, 49 toxic; 88 clusters), with no shared cluster identifiers.

Five frozen outer folds evaluate all six families on identical Development partitions. Within each outer-training set, four `StratifiedGroupKFold` splits grouped by Butina cluster select hyperparameters by mean unweighted AUROC [21]. Outer and inner evaluation groups are disjoint from their corresponding training groups. Data-dependent scaling is fitted on each CV training partition only. Model seed 20260824 and inner split seeds $20260824 + f$, for outer fold f, are preserved. Supplementary Tables D2–D4 and released membership records document class distributions, searches, and final selections. Cluster separation applies to the evaluation folds; the MLP's separate native stopping reservation is described below.

C. Original Classical and Quantum Models

The original comparison comprises logistic regression (LR), Random Forest (RF), multilayer perceptron (MLP), RBF-SVC, Product QK-SVC, and IQP-ZZ QK-SVC. LR uses `class_weight=balanced`, RF uses `balanced_subsample`, and all SVCs use `balanced`. The original MLP uses `StandardScaler` followed by scikit-learn 1.9.0 `MLPClassifier`, ReLU, Adam, `max_iter=2000`, and native early stopping with a stratified 10% molecule reservation. It has no explicit class or sample weights. Repository history establishes that sample weights were omitted when the MLP was introduced; it contains no contemporaneous rationale explaining why. The recorded library supports weighted fitting, so a library restriction is not an established cause. This asymmetry remains a limitation of the original six-family comparison.

The three kernel models use standardized X10 and precomputed-kernel SVCs. "Matched RBF" means shared representation, preprocessing, classifier, and evaluation protocol; the predictive search does *not* force equal bandwidths. Both quantum maps use exact noiseless NumPy statevectors, with no shots or device execution. For standardized z, Product is the separable map

$$|\phi_P(z)\rangle = \bigotimes_{i=1}^{10} R_Y(sz_i)|0\rangle. \tag{1}$$

Its fidelity has the exact classical expression

$$K_P(z, z') = \prod_{i=1}^{10} \cos^2\left(\frac{s(z_i - z_i')}{2}\right). \tag{2}$$

For small angle differences,

$$K_P(z, z') \approx \exp\left[-\frac{s^2}{4}\|z - z'\|_2^2\right], \quad \gamma_{\text{eff}} = s^2/4. \tag{3}$$

Product has no entangling layer. IQP-ZZ starts from $|+\rangle^{\otimes 10}$ and applies a single diagonal phase layer,

$$|\phi_I(z)\rangle = \exp\left[i\left(\sum_i a_i Z_i + \lambda \sum_{i=1}^{9} a_i a_{i+1} Z_i Z_{i+1}\right)\right]|+\rangle^{\otimes 10}, \tag{4}$$

where $a_i = sz_i$ and $\lambda = 1$, with no final Hadamard layer. Both Gram matrices use $|\langle\phi(z)|\phi(z')\rangle|^2$; the implemented definitions, rather than a generic feature-map name, specify the experiments.

D. Retrospective MLP Class-Weight Sensitivity

We preserve the unweighted baseline and add two analyses. For each MLP training fit, class counts from that fit alone determine

$$w_c = \frac{N_{\text{fit}}}{2n_{c,\text{fit}}}, \qquad w_i = w_{y_i}, \quad c \in \{0, 1\}. \tag{5}$$

Weights are passed only to MLP fitting. Scaling and evaluation metrics remain unweighted; no resampling is used. No CV evaluation, holdout, or CR8 labels enter the class counts.

Analysis A fixes each outer fold's hyperparameters, and the final configuration, to those selected by the original MLP. It retains the same inputs, partitions, initializations, preprocessing, and stopping settings. Analysis B repeats the full original search under weighting: hidden layers (32) or $(32, 16)$, $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}\}$, and initial learning rate $\{0.0003, 0.001\}$, giving 12 candidates at the same five-outer/four-inner budget. Mean inner AUROC selects the first highest-scoring `ParameterGrid` candidate in an exact tie. A addresses the direct effect of native weighted training; B addresses the performance of a balanced MLP with full selection. B is not interpreted as isolating balancing at fixed hyperparameters.

Native stopping is preserved in both variants. Its stratified reservation is by molecule, not cluster, and the scaler is fitted before that reservation. Weights use the enclosing CV training partition, including its stopping reserve. In the installed implementation, weighted fitting also weights stopping accuracy and divides the loss gradient and L2 penalty by each minibatch's sum of weights [22]. Thus A includes these native interactions; it does not hold the stopping trajectory or effective batch penalty invariant. The architecture, optimizer, batch-size rule, tolerance, and stopping criterion are otherwise unchanged. Fit-level weights, training memberships, scaler statistics, stopping histories, warnings, and seeds are recorded in Supplementary Tables M1–M7 and the audit bundle.

E. Thresholds, Metrics, and Statistical Scope

For each outer-training partition, we generate complete out-of-fold (OOF) scores using the selected configuration and its four inner folds. Each variant's threshold maximizes MCC, breaking ties by higher balanced accuracy (BA), then smaller absolute threshold, then smaller numerical threshold. Candidate thresholds are the unique OOF scores plus a value just above their maximum; a score at or above the selected threshold is positive. LR/RF/MLP use positive-class probabilities and SVCs use `decision_function`. No external threshold optimization is performed.

For final models, selection uses the five frozen Development folds. The selected configuration's OOF scores on Development determine its threshold under the same rule, followed by one fit on all Development. These selection OOF metrics are not an additional unbiased performance estimate. Final configurations, thresholds, and artifacts are frozen before downstream scoring. The historical holdout had already been inspected during earlier exploration; neither that original exposure nor potential influence on upstream choices can be removed by a later freeze. The new MLP extension also follows observation of holdout and CR8 results and is explicitly retrospective.

AUROC is primary; AUPRC is computed as unweighted average precision, not trapezoidal integration. Both are reported as mean $\pm$ sample SD across outer folds. SD describes partition variability, not a confidence interval. BA and MCC use pooled nested OOF decisions at fold-specific thresholds. We also report sensitivity, specificity, and confusion counts. Paired contrasts use identical outer folds; overlapping training sets limit five-fold inference. The project's additional analysis uses 2,000 paired Butina-cluster bootstrap resamples of fixed pooled OOF predictions, seed 20260824. Its descriptive intervals condition on fitted scores, exclude retraining/selection uncertainty, and are not multiplicity-adjusted tests. Pooled score calibration can differ across folds, so its AUROC/AUPRC contrasts need not equal mean fold contrasts. The original fixed-configuration y-randomization used 200 Development label permutations [23]; it did not repeat nested selection and has not been extended to the new MLP variants.

F. Kernel Geometry, Domain, and External Challenge

Post hoc geometry uses Gram matrices for all 712 Development molecules, standardized by a Development-fitted scaler, at the final selected kernel parameters: RBF $\gamma = 0.03$, Product $s = 0.25$, and IQP-ZZ $s = 0.125$. No labels enter matrix reconstruction, although the parameters were previously selected by supervised CV. We compute centered kernel alignment (CKA) [24], entropy-based effective rank of the uncentered Gram spectrum, and off-diagonal distributions. The separate width control uses RBF $\gamma = 0.015625$, derived from Eq. (3), and is a geometric diagnostic rather than an additional trained predictor.

Two Development-derived applicability domains (ADs) characterize support [25]. The structural criterion is maximum Morgan/Tanimoto similarity with cutoff equal to the fifth percentile of Development leave-one-out maxima, $\tau_S = 0.223134$. The descriptor criterion is mean distance to five neighbors in standardized X10, with cutoff at the 95th leave-one-out percentile, $\tau_d = 2.198975$. The query itself is excluded from Development neighbor calculations. Percentiles are domain conventions, not performance-tuned cutoffs.

TABLE I
ORIGINAL DEVELOPMENT COMPARISON. AUROC/AUPRC: FIVE OUTER-FOLD MEANS $\pm$ SAMPLE SD; BA/MCC: POOLED NESTED OOF DECISIONS AT INNER-OOF-SELECTED THRESHOLDS. ORIGINAL MLP IS UNWEIGHTED.

Model	AUROC	AUPRC	BA	MCC
Random Forest	0.748 ± 0.057	0.657 ± 0.097	0.680	0.468
Logistic Regression	0.733 ± 0.104	0.660 ± 0.124	0.678	0.446
Product QK	0.722 ± 0.108	0.628 ± 0.126	0.658	0.397
IQP-ZZ QK	0.713 ± 0.106	0.645 ± 0.112	0.655	0.400
Matched RBF	0.711 ± 0.108	0.640 ± 0.107	0.663	0.405
MLP original	0.677 ± 0.124	0.596 ± 0.138	0.654	0.428

CR8 consists of eight active ingredients selected from the Costa Rican Servicio Fitosanitario del Estado registry associated with Executive Decree 43838–MAG–S–MINAE [26]. The recorded canonical-SMILES/full-InChIKey check found no overlap with BeeQ893 or ApisTox. This establishes structural independence for the panel, not independence of every downstream interpretation. Its six non-toxic and two toxic compounds support case-level analysis. Nearest toxic and non-toxic Development neighbors, measured separately in structural and descriptor spaces, provide post hoc context only. They do not determine thresholds or selections.

III. RESULTS AND DISCUSSION

A. Development Performance on X10

The original comparison is retained in Table I. RF has the highest mean AUROC, 0.748, while LR has the highest mean AUPRC, 0.660. These averages do not establish uniform model dominance across molecular partitions. The fixed quantum kernels are numerically close to RBF, and variability across folds is substantial (Fig. 2). The original MLP yields AUROC 0.677 ± 0.124 and AUPRC 0.596 ± 0.138; its pooled nested OOF AUROC is 0.668. These estimands must not be confused with final-selection CV performance.

The original y-randomization controls have mean AUROCs near 0.50, whereas real nested OOF scores retain predictive signal (Supplementary Table D5). No randomized score among 200 permutations exceeds its corresponding real score. This finite count should not be reported as a zero probability or as validation of a fully selected modeling pipeline: the control freezes configurations.

B. Balanced MLP: Limited Discrimination Change

Before interpretation, the unweighted MLP was reproduced: selected configurations and predictions for all outer folds, the historical holdout, and CR8 agree with the archived campaign to a maximum absolute score difference of 1.89×10^{-15} (tolerance 10^{-10}). An independent complete rerun reproduces all 18 tabular outputs exactly. The additional analysis contains

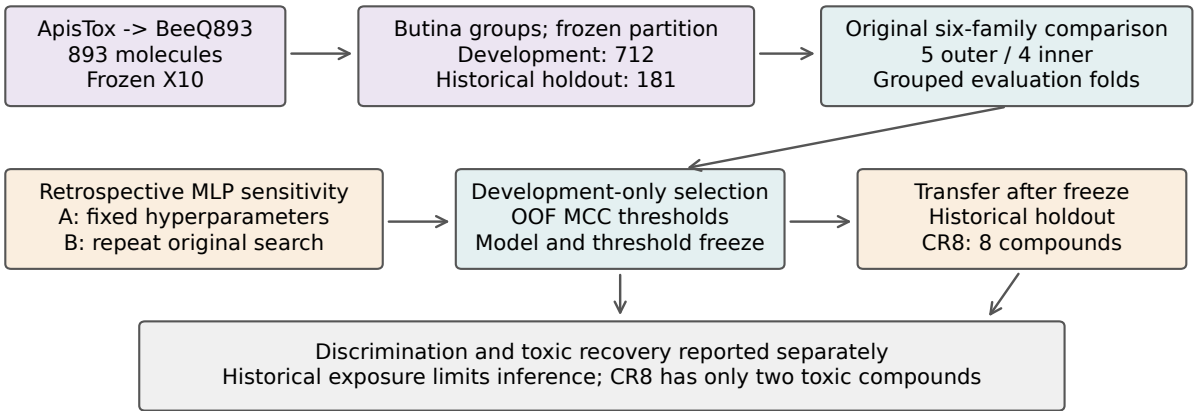


Fig. 1. Frozen modeling protocol and the additional retrospective MLP analysis. Cluster-disjoint nested evaluation and Development-only thresholds separate selection from scoring; earlier exposure of the historical holdout remains an inferential limitation.

TABLE II
ADDITIONAL RETROSPECTIVE MLP DEVELOPMENT ANALYSES. A FIXES ORIGINAL SELECTED HYPERPARAMETERS; B REPEATS THE ORIGINAL SEARCH UNDER WEIGHTING. METRIC AGGREGATION IS THE SAME AS TABLE I.

Variant	AUROC	AUPRC	BA	MCC
Original	0.677 ± 0.124	0.596 ± 0.138	0.654	0.428
Balanced A	0.688 ± 0.100	0.609 ± 0.119	0.638	0.344
Balanced B	0.679 ± 0.100	0.594 ± 0.112	0.632	0.338

693 fits, all stopping before 2,000 epochs (12–77), with no captured training warnings. This verifies the configured stopping behavior and finite computations, not a global or stationary optimum.

Analysis A increases mean outer AUROC by 0.0113 to 0.688 and AUPRC to 0.609; B gives AUROC 0.679 and AUPRC 0.594 (Table II). Descriptive pooled AUROC bootstrap intervals include zero for both changes (A: $[-0.0242, 0.0356]$; B: $[-0.0369, 0.0258]$). Retuned B has a negative pooled AUPRC contrast, -0.0474, with interval $[-0.0989, -0.0009]$, despite its much smaller mean-fold change. This negative result further precludes a general claim of improved discrimination.

At variant-specific MCC thresholds, pooled toxic recovery is 75/222 originally, 79/222 in A, and 75/222 in B. False positives rise from 15/490 to 39/490 and 36/490; MCC falls from 0.428 to 0.344 and 0.338. Sensitivity improves in four folds but drops from 12/37 to 1/37 in fold 3 in both balanced variants (Fig. 3). Therefore sensitivity does not improve consistently. The original weighting omission affects decisions, but the sensitivity analysis does not establish it as the explanation for the MLP's overall lower discrimination or its transfer gap. Both balanced means remain below those of the other five families.

C. Historical Transfer: Toxic Recovery and Its Cost

The original-model holdout AUROCs span 0.462–0.706, including the MLP. RBF is 0.706, Product 0.697, and IQP-ZZ 0.696. Original sensitivities are 0.082–0.265, with specificity 0.947–0.985 (Table III). Even models retaining moderate ranking performance identify few toxic compounds at thresholds selected on Development. The MLP has the largest discrimination gap, from mean outer AUROC 0.677 to holdout AUROC 0.462 and AUPRC 0.302.

With separately frozen thresholds 0.524583, 0.536224, and 0.528334 for original/A/B, MLP recovery increases from 4/49 to 11/49 and 13/49. False positives increase from 4 to 16 and 22, and specificity decreases from 0.970 to 0.879 and 0.833. AUROC remains 0.481–0.482 and AUPRC 0.329–0.332 in the balanced variants. The most marked change is therefore in positive decisions and their specificity cost, while discrimination remains weak. Because the fitted scores and thresholds both change, this is not attributed to moving a threshold alone. These retrospective observations cannot establish deployment performance or confirm that balancing closes a generalization gap.

The historical partition has disjoint molecules and clusters, but its earlier inspection could have influenced descriptor design, curation, or other upstream choices. A final freeze protects against new downstream retuning; it cannot undo those possible pathways. These results consequently provide exploratory transfer evidence, rather than a pristine confirmatory test. CR8 avoids the same structural overlap, but its sample size gives much weaker precision for population performance. The two sets cannot be ordered on trustworthiness using a single label such as "external."

D. Quantum–RBF Contrasts and Similarity Geometry

Mean paired AUROC contrasts are $+0.0110$ for Product–RBF, $+0.0024$ for IQP-ZZ–RBF, and $+0.0086$ for Product–IQP-ZZ; their signs vary across folds (Fig. 4). The holdout Product–RBF and IQP-ZZ–RBF contrasts are -0.0083 and

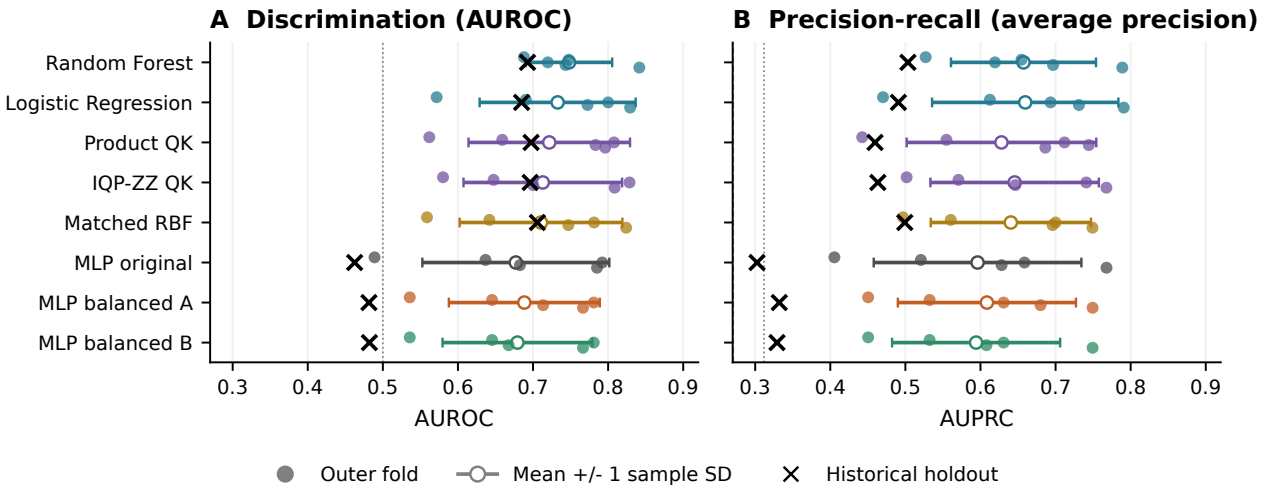


Fig. 2. Development and historical-holdout discrimination. Points show the five outer folds, open circles the means, and bars one sample SD; crosses show historical-holdout scores. Added MLP A/B rows are retrospective sensitivity analyses. AUROC's reference is 0.5; the AUPRC prevalence references differ between Development (0.312) and holdout (0.271). SD is not a confidence interval.

TABLE III
HISTORICAL HOLDOUT ($n = 181$: 49 TOXIC, 132 NON-TOXIC) AT FROZEN DEVELOPMENT-SELECTED THRESHOLDS. THE SIX ORIGINAL RESULTS ARE PRESERVED; BALANCED A/B ARE ADDITIONAL RETROSPECTIVE ANALYSES. NO SAMPLE WEIGHTING IS USED IN EVALUATION.

Model	AUROC	AUPRC	BA	MCC	Sens.	Spec.	TP	FN	TN	FP
Random Forest	0.693	0.503	0.618	0.358	0.265	0.970	13	36	128	4
Logistic Regression	0.685	0.491	0.555	0.178	0.163	0.947	8	41	125	7
Product QK	0.697	0.460	0.533	0.165	0.082	0.985	4	45	130	2
IQP-ZZ QK	0.696	0.463	0.529	0.136	0.082	0.977	4	45	129	3
Matched RBF	0.706	0.499	0.543	0.200	0.102	0.985	5	44	130	2
MLP original	0.462	0.302	0.526	0.111	0.082	0.970	4	45	128	4
MLP balanced A	0.481	0.332	0.552	0.129	0.224	0.879	11	38	116	16
MLP balanced B	0.482	0.329	0.549	0.111	0.265	0.833	13	36	110	22

-0.0096. Fixed-score cluster-bootstrap pooled AUROC contrasts are $+0.0017$, -0.0014, and $+0.0032$, respectively, with all descriptive intervals including zero. Thus the tested replacement does not yield a persistent predictive improvement. These are observations about these maps and X10, not a general impossibility result for quantum kernels.

Reconstruction covers 253,116 unique Development pairs. CKA is 0.982454 for Product–RBF, 0.985072 for IQP-ZZ–RBF, and 0.999506 for Product–IQP-ZZ. At Product $s = 0.25$, Eq. (3) implies $\gamma_{\text{eff}} = 0.015625$. Product's effective rank is 3.086, compared with 5.517 for the selected RBF ($\gamma = 0.03$) and 3.078 for the width-control RBF. Product/control CKA is 0.999794. This provides an explicit bandwidth explanation for much of the apparent spectral difference; the small-angle approximation does not imply global identity. The exact Product formula reproduces the simulated matrix within 3.89×10^{-15}.

These results connect to, but do not establish, the concentration mechanism studied by Thanasilp et al. [4]. That mechanism concerns kernel variation and measurement resources as qubit number grows. Here we use one fixed qubit count and exact overlaps; Product and IQP-ZZ off-diagonal SDs remain 0.165 and 0.167. High CKA is a finite-cohort similarity diagnostic, not the geometric-difference quantity in the theoretical bound [3], a test of exponential concentration, or evidence of a barren plateau. The stronger explanation supported here is the analytic near-Gaussian form and bandwidth-controlled alignment.

Concrete alternatives include careful bandwidth selection [27], projected kernels based on reduced-state observables [3], and training feature-map parameters by kernel-target alignment [28]. These alter different aspects of the kernel and require nested selection and classical controls. Trainability itself is not a guaranteed remedy, because alignment optimization can also concentrate under unfavorable embeddings [4]. They are candidate follow-up studies, not methods tested by BeeQ.

E. CR8 Decisions and Applicability Domain

In the original six-family comparison, all models miss Ethiprole and Isocycloseram, the two toxic CR8 compounds. The other five models classify all six non-toxic compounds correctly, while the original MLP falsely flags Florylpicoxamid. The retrospective balanced variants both recover Isocycloseram, still miss Ethiprole, and additionally flag Fluoxapiprolin; the Florylpicoxamid false positive remains

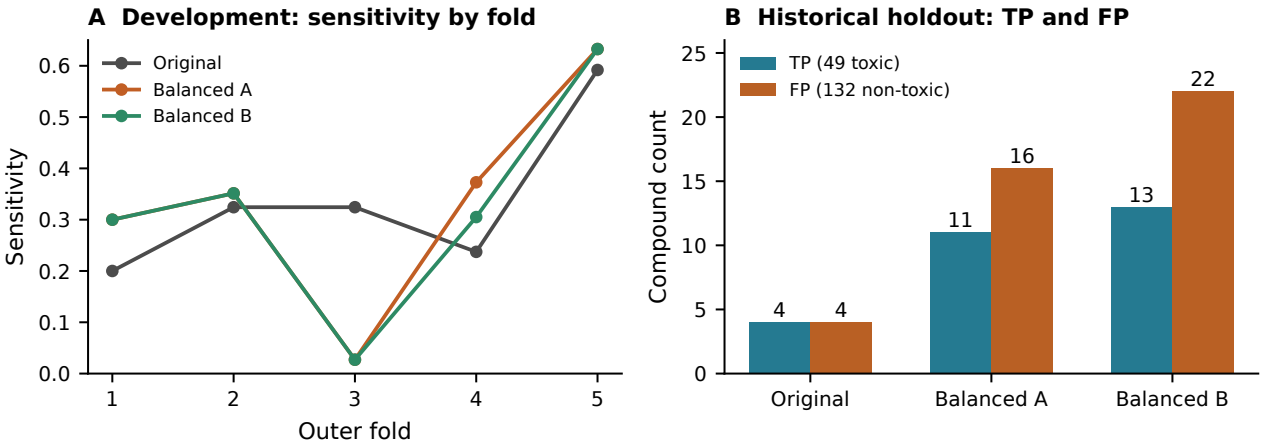


Fig. 3. Threshold consequences of balancing. (A) Sensitivity on identical outer folds at separately selected inner-OOF thresholds; the fold-3 decline prevents a claim of consistent improvement. (B) Historical-holdout true positives and false positives at frozen final thresholds. A uses original hyperparameters; B repeats selection.

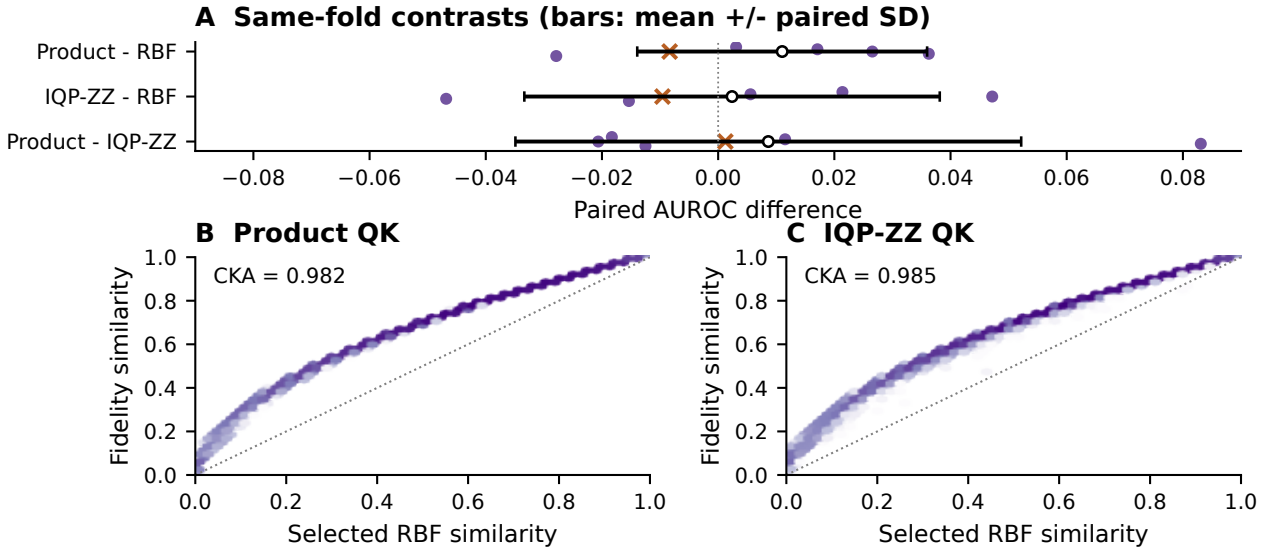


Fig. 4. Predictive and geometric kernel comparisons. (A) Paired outer-fold AUROC contrasts; open circles and bars show mean ± paired sample SD, and crosses the historical holdout. (B,C) Density of the 253,116 unique Development molecular pairs for the selected RBF and each fidelity kernel. Reconstruction is post hoc at supervised-selected parameters. High CKA does not establish mathematical identity or exponential concentration.

(Fig. 5). Toxic recovery thus depends on the MLP training variant. Compound scores, decisions, thresholds, and aggregate descriptive metrics are provided in Supplementary Table M5. MLP AUROC/AUPRC decrease from 0.833/0.583 originally to 0.667/0.450 in A and 0.583/0.367 in B, illustrating that recovering a toxic case at a threshold does not imply better ranking. One changed decision among two positives cannot establish regional or population sensitivity.

Seven CR8 compounds are inside both Development-derived ADs. Fluoxapiprolin is structurally inside but outside the descriptor domain. Both toxic compounds are inside both domains, irrespective of their prediction. The Development leave-one-out structural and descriptor support measures have Spearman $\rho = -0.313$ and describe different aspects of proximity. Ethiprole's nearest toxic structural neighbor has Tanimoto 0.740, whereas Isocycloseram's has 0.233. These examples show that domain membership is not a guarantee of correctness. They do not identify a missing toxicological mechanism or prove that representation expansion would correct the errors. The recorded labels remain fixed; supporting endpoint sources include the acute Ethiprole study [29] and the Japanese Isocycloseram assessment [30].

F. Context from Published ApisTox Benchmarks

Table IV adds quantitative context across fingerprints, graph kernels, a GNN, and pretrained embeddings [9]. The published study uses 1,035 molecules and MaxMin or approximated-time

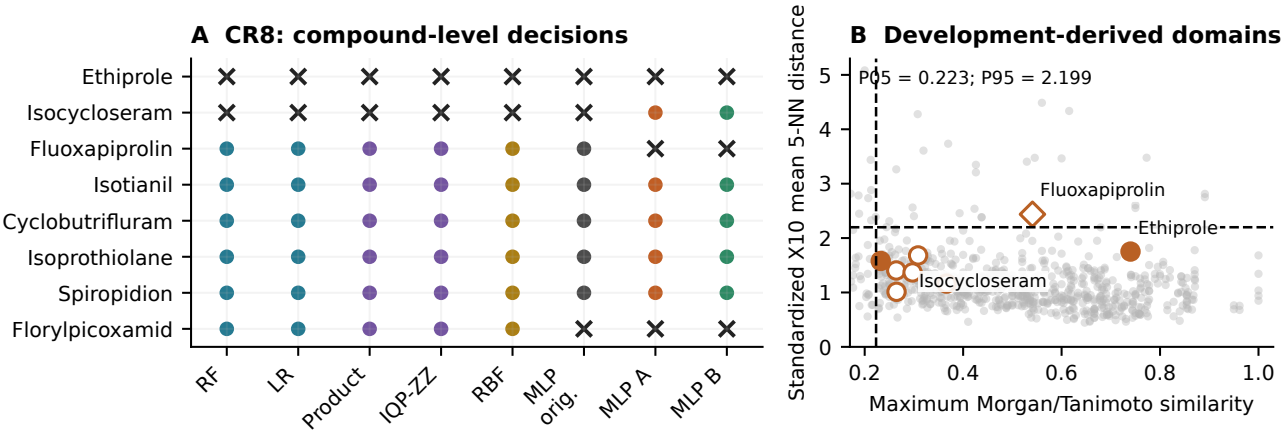


Fig. 5. CR8 case-level decisions and chemical support. (A) Circles indicate correct decisions and crosses errors; the first two rows are toxic. The last two columns are additional retrospective MLP analyses. (B) Development leave-one-out support (gray) and CR8 queries (orange; filled markers toxic), with cutoffs derived only from Development. Both toxic queries remain inside both domains; Fluoxapiprolin is outside the descriptor domain.

TABLE IV
PUBLISHED APISTOX CONTEXT [9], TABLES 1–2 AND APPENDIX B, AND BEEQ. AUROC SHOWN ON THE 0–1 SCALE. BENCHMARK SDS REFLECT 50 SEEDS FOR RF/GNNS; DETERMINISTIC ROWS HAVE NO SD. BEEQ SD REFLECTS OUTER FOLDS. NR: AUPRC NOT REPORTED IN THOSE SOURCE TABLES. THESE ARE NOT MATCHED PERFORMANCE COMPARISONS.

Representation/model	MaxMin AUROC	Time AUROC	AUPRC
Avalon / RF	0.761 ± 0.013	0.773 ± 0.010	NR
ECFP / RF	0.717 ± 0.014	0.782 ± 0.012	NR
WL-OA / SVM	0.840	0.781	NR
AttentiveFP / GNN	0.752 ± 0.019	0.708 ± 0.023	NR
Mol2Vec / LR	0.766	0.691	NR
BeeQ X10 / RF	0.748 ± 0.057 (nested)		0.657 ± 0.097

20% test sets; BeeQ uses 893 curated molecules and nested cluster folds. Its seed variability and BeeQ's fold variability estimate different uncertainties. AUPRC is not available in those source tables. The numerical comparison neither ranks these incompatible protocols nor establishes that X10 is equivalent to richer representations.

IV. LIMITATIONS AND NEXT EXPERIMENTS

The principal limit is the fixed representation. X10 supplies a controlled ten-input comparison, but neither its dimensionality nor its descriptor composition is validated as optimal for acute toxicity. Without a richer representation evaluated on the same molecules and nested folds, we cannot separate an information bottleneck from limitations of the learning families. Fingerprint or expanded-descriptor sensitivity, followed by controls for kernel geometry and toxic recovery, is therefore needed before generalizing the negative quantum result beyond X10. Such an experiment is not included in this revision, and no improvement from it is asserted.

The original MLP's lack of balancing is retained as historical context. Weighted reruns address that specific asymmetry and its effect under the original protocol, but native early stopping is still not cluster disjoint: its gradient-training and stopping subsets share 14–32 clusters. The scaler also includes the native stopping reserve, while excluding the CV evaluation fold. Preserving this behavior supports a fair balancing comparison but leaves the stopping rule as a methodological limitation. A separately controlled group-based stopping study could address it; none is claimed here.

The previously inspected holdout has limited confirmatory status, and all added downstream evaluations are retrospective. CR8 has independent structures but only two toxic cases; its errors and corrected decisions are qualitative cases rather than an estimate of regional prevalence, sensitivity, or regulatory suitability. The endpoint pools exposure routes and assays, and excludes chronic, sublethal, and field-exposure effects. MCC-selected thresholds also encode a particular tradeoff; they have not been optimized for a prospective toxic-screening utility or calibrated under a new deployment distribution.

Finally, the study uses exact classical simulation and no quantum device measurements. It does not demonstrate computational speedup, hardware feasibility, or a general quantum advantage. Product's exact closed form further makes its classical evaluability explicit. Broader representations, projected or trained kernels, and concentration diagnostics across qubit counts would answer different questions and require new experiments with appropriate controls. More circuit complexity alone is not supported as a remedy by these results.

V. CONCLUSIONS

BeeQ establishes two bounded findings. First, with X10 and a shared molecular-cluster protocol, the tested fixed fidelity kernels show no persistent predictive advantage over RBF and induce highly aligned finite-cohort geometries; the Product width control largely removes its apparent effective-rank difference. Second, Development discrimination does not ensure recovery of toxic compounds at frozen thresholds beyond Development. The historical holdout exposes low sensitivity,

while the tiny CR8 panel provides concrete examples of errors despite domain membership.

The retrospective MLP correction increases toxic recovery in the holdout and detects Isocycloseram in CR8, but adds false positives, does not consistently improve Development sensitivity, and does not resolve weak holdout discrimination. These results qualify the original MLP comparison without changing the quantum–RBF evidence. They support cautious research use of this controlled protocol as an exploratory computational New Approach Methodology (NAM), rather than a validated screening or regulatory tool. The remaining priority is a prospectively evaluated, representation-sensitive study of toxic recovery and its false-positive cost; the present data do not determine which representation or kernel adaptation will succeed.

Data and Code Availability

The BeeQ repository is https://github.com/andresquesadag/BeeQ. Original results remain in the retained final nested campaign; additional MLP results, source/input hashes, split memberships, selected parameters, OOF scores, fit histories, and frozen external predictions are stored separately in `results/mlp_sensitivity/full_20261009_v2`. The revision sources, table/figure generator, provenance ledger, and build checks are in `manuscript/final_revision`. The supporting supplement maps every quantitative result to its artifact. Model evaluation is reproducible from the released frozen X10 tables; regeneration of supplied in-house descriptors requires their generating workflows. No original result or model has been overwritten.

References

[1] M. Schuld and N. Killoran, "Quantum machine learning in feature Hilbert spaces," *Phys. Rev. Lett.*, vol. 122, p. 040504, 2019, doi: 10.1103/PhysRevLett.122.040504.

[2] V. Havlíček et al., "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, pp. 209–212, 2019, doi: 10.1038/s41586-019-0980-2.

[3] H.-Y. Huang et al., "Power of data in quantum machine learning," *Nat. Commun.*, vol. 12, p. 2631, 2021, doi: 10.1038/s41467-021-22539-9.

[4] S. Thanasilp, S. Wang, M. Cerezo, and Z. Holmes, "Exponential concentration in quantum kernel methods," *Nat. Commun.*, vol. 15, p. 5200, 2024, doi: 10.1038/s41467-024-49287-w.

[5] A. M. Smaldone et al., "Quantum machine learning in drug discovery: Applications in academia and pharmaceutical industries," *Chem. Rev.*, vol. 125, pp. 5436–5460, 2025, doi: 10.1021/acs.chemrev.4c00678.

[6] A.-M. Klein et al., "Importance of pollinators in changing landscapes for world crops," *Proc. R. Soc. B*, vol. 274, pp. 303–313, 2007, doi: 10.1098/rspb.2006.3721.

[7] P. Yang, E. A. Henle, X. Z. Fern, and C. M. Simon, "Classifying the toxicity of pesticides to honey bees via support vector machines with random walk graph kernels," *J. Chem. Phys.*, vol. 157, p. 034102, 2022, doi: 10.1063/5.0090573.

[8] J. Adamczyk, J. Poziemski, and P. Siedlecki, "ApisTox: A new benchmark dataset for the classification of small molecules toxicity on honey bees," *Sci. Data*, vol. 12, p. 5, 2025, doi: 10.1038/s41597-024-04232-w.

[9] J. Adamczyk, J. Poziemski, and P. Siedlecki, "Evaluating machine learning models for predicting pesticide toxicity to honey bees," *Ecotoxicol. Environ. Saf.*, vol. 312, p. 119869, 2026, doi: 10.1016/j.ecoenv.2026.119869. Open author version: https://arxiv.org/html/2503.24305v4.

[10] A. Tropsha, "Best practices for QSAR model development, validation, and exploitation," *Mol. Inform.*, vol. 29, pp. 476–488, 2010, doi: 10.1002/minf.201000061.

[11] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, p. 91, 2006, doi: 10.1186/1471-2105-7-91.

[12] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *JMLR*, vol. 11, pp. 2079–2107, 2010.

[13] RDKit contributors, "RDKit: Open-source cheminformatics software," version 2026.03.5, 2026. https://www.rdkit.org.

[14] D. Fourches, E. Muratov, and A. Tropsha, "Trust, but verify: On the importance of chemical structure curation in cheminformatics and QSAR modeling research," *J. Chem. Inf. Model.*, vol. 50, pp. 1189–1204, 2010, doi: 10.1021/ci100176x.

[15] K. Mansouri et al., "Free and open-source QSAR-ready workflow for automated standardization of chemical structures in support of QSAR modeling," *J. Cheminform.*, vol. 16, p. 19, 2024.

[16] U.S. EPA, Health Canada PMRA, and California DPR, *Guidance for Assessing Pesticide Risks to Bees*, June 19, 2014.

[17] S. A. Wildman and G. M. Crippen, "Prediction of physicochemical parameters by atomic contributions," *J. Chem. Inf. Comput. Sci.*, vol. 39, pp. 868–873, 1999, doi: 10.1021/ci990307l.

[18] L. A. Zumbado Silva and W. J. Zamora Ramírez, "LiPHEX: A QSPR framework for predicting n-hexadecane/water logP values," *ChemRxiv*, preprint, April 2026, doi: 10.26434/chemrxiv.15002074/v1.

[19] D. Rogers and M. Hahn, "Extended-connectivity fingerprints," *J. Chem. Inf. Model.*, vol. 50, pp. 742–754, 2010, doi: 10.1021/ci100050t.

[20] D. Butina, "Unsupervised data base clustering based on Daylight's fingerprint and Tanimoto similarity: A fast and automated way to cluster small and large data sets," *J. Chem. Inf. Comput. Sci.*, vol. 39, pp. 747–750, 1999, doi: 10.1021/ci9803381.

[21] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011. Installed version: 1.9.0.

[22] Scikit-learn developers, "MLPClassifier API documentation," version 1.9. https://scikit-learn.org/1.9/modules/generated/sklearn.neural_network.MLPClassifier.html. Installed implementation and source hashes are recorded with the sensitivity results.

[23] C. Rücker, G. Rücker, and M. Meringer, "y-randomization and its variants in QSPR/QSAR," *J. Chem. Inf. Model.*, vol. 47, pp. 2345–2357, 2007, doi: 10.1021/ci700157b.

[24] C. Cortes, M. Mohri, and A. Rostamizadeh, "Algorithms for learning kernels based on centered alignment," *JMLR*, vol. 13, pp. 795–828, 2012.

[25] OECD, *(Q)SAR Assessment Framework: Guidance for the Regulatory Assessment of (Quantitative) Structure–Activity Relationship Models and Predictions*, 2nd ed., Series on Testing and Assessment no. 405, 2024.

[26] Servicio Fitosanitario del Estado, "Ingredientes activos grado técnico aprobados bajo decreto n. 43838–MAG–S–MINAE," Costa Rica, Sept. 9, 2026. https://www.sfe.go.cr/DocsStatusRegistro/IAGT_aprobados_bajo_Decreto_43838.pdf. Current verification; not an archived panel-selection snapshot.

[27] R. Shaydulin and S. M. Wild, "Importance of kernel bandwidth in quantum machine learning," *Phys. Rev. A*, vol. 106, p. 042407, 2022, doi: 10.1103/PhysRevA.106.042407.

[28] T. Hubregtsen et al., "Training quantum embedding kernels on near-term quantum computers," *Phys. Rev. A*, vol. 106, p. 042431, 2022, doi: 10.1103/PhysRevA.106.042431.

[29] T. Cang et al., "Acute toxicities and preliminary risk evaluation of chiral ethiprole to three non-target organisms," *Chin. J. Pestic. Sci.*, vol. 18, pp. 65–70, 2016, doi: 10.16801/j.issn.1008-7303.2016.0007.

[30] Ministry of Agriculture, Forestry and Fisheries of Japan, "Isocycloseram pesticide honey-bee impact assessment report," Agricultural Materials Council, June 5, 2024, in Japanese. https://www.maff.go.jp/j/council/sizai/nouyaku/attach/pdf/mitubati_13-14.pdf.